

AI Customer Support Ticket Classifier

GenAI / NLP / AI Automation Track

Project starts	Saturday, August 22, 2026
Project work period	August 22 - September 1, 2026
Project upload days	Monday, August 31 & Tuesday, September 1, 2026
Final submission deadline	Tuesday, September 1, 2026
Feedback, corrections & video	Wednesday, September 2 - Friday, September 4, 2026

1. Business Scenario

You have joined the AI/NLP team of a financial-services company. The customer-support team receives many text requests every day. Agents currently read each request and manually route it to the appropriate support category. Your task is to build a small, interpretable NLP prototype that predicts the intent of a new customer message and supports automatic ticket routing.

This project is not a chatbot or RAG project. The goal is to learn text classification, evaluation, error analysis and business-aware routing.

2. Public Dataset

BANKING77 is a public English-language benchmark for fine-grained online-banking intent classification. It contains 13,083 customer-service queries across 77 intents (10,003 train and 3,080 test examples). For Project 01, use a manageable subset of 6-8 intents.

Main dataset: <https://huggingface.co/datasets/PolyAI/banking77>

Original PolyAI repository: <https://github.com/PolyAI-LDN/task-specific-datasets>

Original train CSV:

https://github.com/PolyAI-LDN/task-specific-datasets/blob/master/banking_data/train.csv

Research paper: <https://arxiv.org/abs/2003.04807>

Dataset license: CC BY 4.0.

Recommended subset

card_arrival | card_not_working | cash_withdrawal_not_recognised | declined_card_payment | lost_or_stolen_card | transaction_charged_twice | transfer_not_received_by_recipient | cash_withdrawal_charge

3. What You Should Know / Read

Python: functions, lists/dictionaries, pandas filtering/groupby, basic plotting.

NLP: token, text classification, Bag of Words, TF-IDF, n-grams, stop words.

ML: train/test split, Logistic Regression, overfitting, Precision, Recall, F1-score, Macro-F1, confusion matrix.

Scientific workflow: baseline comparison, data leakage, reproducibility, error analysis and limitations.

Required: Python, Jupyter/Colab, Git/GitHub, pandas, NumPy, Matplotlib, scikit-learn. Optional: Streamlit.

4. Questions from the Company

Q1 - What are customers contacting us about most often?

Measure count and proportion by intent. Create a horizontal bar chart. Explain which support categories appear to carry the largest workload and the operational implication.

Q2 - Do message length and structure differ by intent?

Create character-length and word-count features. Compare distributions with suitable plots and explain whether these features appear informative.

Q3 - Can a simple keyword system route tickets?

Create a small rule-based baseline for obvious phrases. Evaluate where it succeeds and fails. Explain why ML may be needed.

Q4 - Can classical NLP predict customer intent?

Build a reproducible TF-IDF + Logistic Regression classifier. Fit text transformations only on training data.

Q5 - Which intents does the model understand well, and which does it confuse?

Report Precision, Recall, F1, Macro-F1 and a confusion matrix. Identify commonly confused intent pairs and discuss likely linguistic reasons.

Q6 - Are all errors equally costly?

Select 2-3 business-critical intents and discuss why false negatives/misrouting may matter more for them and which metric deserves attention.

Q7 - How should the system handle a new message?

Create a reusable function returning predicted intent and, where supported, an interpretable probability/confidence.

Q8 - When should the system defer to a human?

Create a low-confidence human-review rule. Justify the threshold and discuss its limitations.

5. Required Visualizations

At least five useful figures: **intent distribution**; **message/word length by intent**; **top TF-IDF terms for selected intents**; **confusion matrix**; **per-class F1-score**. Every figure must include a written interpretation.

6. Recommended Workflow

Data understanding → select 6-8 intents → text EDA → train/test strategy → keyword baseline → TF-IDF → Logistic Regression → evaluation → inspect at least 20 errors → prediction function / optional Streamlit app.

7. Scientific & Quality Requirements

No data leakage: TF-IDF must be fit on training data only; a scikit-learn Pipeline is recommended.

Do not report Accuracy alone: include class-level metrics and Macro-F1.

Error analysis: manually inspect at least 20 incorrect predictions.

Reproducibility: code should run cleanly from top to bottom and use a fixed random seed where applicable.

Interpretation: distinguish evidence from assumptions and do not claim the prototype is production-ready.

Not required in Project 01: BERT, Transformers, RAG, LangChain, vector databases or multi-agent systems.

8. Submission Rules - Mandatory

Upload your project on Monday, August 31 or Tuesday, September 1, 2026.

The final submission deadline is Tuesday, September 1, 2026.

Wednesday, September 2 through Friday, September 4 is reserved for feedback, corrections and project-video submission.

ZIP/RAR files are not accepted. Submit one clean, organized folder/repository.

README.md is mandatory.

A code file is mandatory: at least one readable .ipynb and/or .py file.

Recommended Repository Structure

```
customer-support-ticket-classifier/  
  README.md  
  requirements.txt  
  notebooks/analysis.ipynb  
  src/predict.py  
  figures/  
  data/README.md
```

9. README Requirements

Project title; business problem; dataset source/link and license; selected intents; methodology; EDA findings; baseline; TF-IDF + Logistic Regression; evaluation; error analysis; business implications; limitations; repository structure; and instructions for running the code.

10. Final Deliverable

A successful project should show that you can frame a business problem, understand text data, establish a baseline, build a reproducible NLP pipeline, evaluate it correctly, inspect failure cases and explain when automation should defer to a human.

Suggested GitHub title: AI Customer Support Ticket Classifier - Intent Detection & Ticket Routing with NLP